

Predictive Tire Wear Monitoring System

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Abstract—The Predictive Tire Wear Monitoring System introduces a modern approach to improving vehicle safety and efficient tire maintenance through real-time analysis of tire conditions. Unlike conventional methods, which often overlook dynamic factors such as road surfaces, driving styles, and vehicle loads, this system combines data from accelerometers, gyroscopes, vibration sensors, and GPS modules, all managed by an ESP32 microcontroller. Three machine learning algorithms—Support Vector Machine (SVM), Convolutional Neural Network (CNN), and Deep Neural Network (DNN)—were tested to predict tire wear patterns. Among these, DNN demonstrated the highest accuracy and reliability, making it the algorithm of choice for the system. This proactive maintenance strategy reduces tire-related risks, optimizes usage, and lowers costs, while also supporting environmental sustainability. The system is designed to cater to both individual users and fleet operators, offering a scalable, reliable solution for predictive maintenance.

Keywords— Tire Wear, Sensor Fusion, Predictive Maintenance, Accelerometer, GPS, Vibration Sensor, Machine Learning.

I. INTRODUCTION

Monitoring tire health is critical for vehicle safety, especially in India, where road conditions and traffic patterns pose significant challenges to vehicle performance. Poor tire conditions can lead to accidents such as blowouts, traction loss, and increased braking distances. According to the Ministry of Road Transport and Highways (MoRTH), tire-related accidents contribute significantly to road fatalities in India. A report by the Indian Road Safety Campaign estimates that approximately 30% of all road accidents in India are caused by tire-related issues, including worn-out tires, improper tire pressure, and inadequate tread depth. Specifically, tire blowouts and loss of traction due to worn

tread are responsible for a significant portion of these incidents, leading to over 10,000 tire-related accidents annually in India. These accidents result in injuries and fatalities that could be prevented with better tire monitoring [1]. Research suggests that tire blowouts alone contribute to about 15-20% of all road accidents in India, further emphasizing the need for effective tire health monitoring systems.

Current systems, such as Tire Pressure Monitoring Systems (TPMS), focus primarily on monitoring tire pressure but fail to account for other critical factors, like tread depth, vibration levels, and road conditions, which are essential for a comprehensive understanding of tire health and directly impact vehicle safety. In India, where road conditions vary drastically—from smooth highways to poorly maintained rural roads—the need for a holistic tire monitoring system that includes tread depth and vibration sensors is even more critical [2][3]. While advanced sensor arrays and intelligent tire concepts have been proposed, many solutions remain in the experimental phase or confined to controlled environments, lacking robustness for real-world use, especially in India's diverse driving conditions [4][8]. Furthermore, integrating data from multiple sensors, such as accelerometers, vibration sensors, and GPS, continues to pose challenges for effective tire wear analysis in real-world conditions [11][12].

Previous studies on tire wear prediction have focused on laboratory setups and controlled environments, which do not capture real-world driving variability. For example, Han et al. used machine learning to classify tread wear, but their study was limited to controlled tests, reducing its real-world applicability [5]. Similarly, Arora et al. and Zhang et al. used vibration and strain gauge features, but struggled to represent actual driving conditions, especially in regions with diverse road types like India [4][8]. The primary gap in current tire wear research is the inability to effectively integrate multi-sensor data under varying road conditions. While some studies

have applied machine learning models to predict tire wear, they fail to account for factors like vehicle dynamics, driving behaviors, and the wide range of road surfaces encountered in India. The lack of a system that can adapt to these real-world conditions presents a significant research gap.

This study proposes a Predictive Tire Wear Monitoring System that integrates accelerometers, gyroscopes, vibration sensors, and GPS modules with machine learning algorithms. By combining real-time data from multiple sensors, the system analyzes tire movement, road conditions, and driving behavior, ensuring accurate tire wear prediction and timely maintenance alerts, which are crucial for improving safety, particularly in countries like India where road conditions and driving habits vary significantly. Unlike previous approaches, which often rely on controlled environments, this system is designed to function reliably in real-world conditions, addressing critical gaps in tire wear prediction research. It uses a multi-sensor approach for more accurate and comprehensive data analysis, ultimately improving safety and cost efficiency for vehicles on Indian roads.

II. LITERATURE ANALYSIS

A. Reference and Contributions

The Predictive Tire Wear Monitoring System is shaped by several studies that highlight the influence of various factors on tire wear, including vehicle type, tire features, and driving behavior. These insights align with our model's focus on driving behavior and tire wear prediction. Research on sensor-based tire health monitoring guided our decision to use accelerometers, vibration sensors, and GPS for real-time data collection [1-3].

The use of weightless neural networks for tire condition monitoring informed our choice of machine learning

techniques for processing vibration data [4]. Experimental evaluation methods played a key role in guiding our validation process, ensuring that real-road data was incorporated effectively [5]. Additionally, sensor fusion techniques enhanced our system's accuracy by combining data from multiple sensors [6-7].

Accelerometer data analysis techniques were essential in refining our feature extraction process [8]. The integration of machine learning models helped us select the best algorithms for predicting tire wear, while real-time strategies ensured that timely feedback could be provided to users [9]. Methods for tire-road interaction recognition were instrumental in improving the model's understanding of road conditions and tire wear patterns [10-15].

Further studies on driving behavior, printed sensor arrays for tread thickness, and intelligent tire concepts helped refine our sensor-based health monitoring [16-17]. Research also contributed to the improvement of vibration signal processing and low-power system design, which were key aspects of our approach to real-world validation [18-20]. Preprocessing and estimation algorithms directly impacted our model's accuracy, while real-time tire force estimation strategies enabled immediate feedback to users [21-24]. Finally, research on unsupervised learning for tire defect inspection and sensor performance comparison provided insights that guided our machine learning approach. The application of CNNs inspired our deep learning models, while studies on tire pressure and slip angle estimation enhanced our system's predictive capabilities [25-30].

B. Tabulation of Reference Papers

The following table 1 consolidates several papers and classifies them based on their methods and results,

TABLE I.
REFERENCE PAPERS

Reference No.	Methodologies	Key Focus Area	Findings	Limitations	Performance Metrics
[1]	Sensor fusion, accelerometers	Correlation Between Driving Behavior and Tire Wear	Effective correlation between driving behavior and tire wear	Limited to specific vehicle types and driving conditions	High correlation (R^2 value), prediction accuracy
[2]	Printed sensor array	Mapping Tire Tread Thickness Profiles	Provides detailed tread thickness profiles	Limited scalability in diverse environments	Accuracy in profile mapping, signal noise reduction
[3]	Intelligent tire concept	Real-Time Tire Health Monitoring	Continuous, real-time tire health data	Potential issues with sensor reliability in extreme conditions	Real-time monitoring accuracy, data update frequency
[4]	Weightless neural networks, MEMS sensors	Enhanced Prediction Using MEMS Sensors and Neural Networks	Enhanced prediction accuracy using vibration signals	High computational cost, sensor calibration required	Prediction accuracy, training time efficiency
[5]	Machine learning	Tire Tread Wear Detection Using Machine Learning	High accuracy for detecting wear under varying conditions	Limited generalization to diverse tire types and road conditions	Wear detection accuracy, false positive rate
[8]	AI, smart sensors	Road Condition Monitoring with Smart Sensing and AI	Accurately predicts road conditions affecting tire wear	Need for more comprehensive data for diverse scenarios	Road condition prediction accuracy, response time
[9]	Accelerometer data	Surface State	Helps in detecting road	Limited sensor range	Detection accuracy,

	analysis	Identification Using Accelerometer Data	surface conditions affecting tire wear	and sensitivity	sensor data processing time
[13]	Transfer learning, deep neural networks	Enhanced Prediction Accuracy with Transfer Learning	Enhances prediction accuracy with minimal data	Requires significant pre-training data for accuracy	Prediction accuracy, data usage efficiency
[16]	Machine learning, neural networks	Estimation of Tire Wear Based on Vehicle Type and Driving Behavior	Provides an effective model for estimating tire wear based on various factors	Potential overfitting with limited vehicle types	Estimation accuracy, model complexity
[27]	1D Convolutional Neural Networks (CNN)	High-Accuracy Wear Prediction with Time-Series Data	High accuracy in tire wear prediction	Model may not generalize well for different tire types and environments	Prediction accuracy, model training time, test data accuracy

III. MATERIALS AND METHODS

This section outlines the materials and methods for developing the Predictive Tire Wear Monitoring System, including system architecture, sensor integration, and real-time data processing. It covers the sensors, data acquisition, and machine learning algorithms used for tire wear prediction.

A. Dataset

The dataset for the Predictive Tire Wear Monitoring System was obtained from Kaggle [31], containing sensor data from various driving speeds and tire wear conditions including automotive and sensor data repositories. It consists of accelerometer, vibration sensor, and GPS module readings across various driving conditions, with tire wear ranging from 0% (new) to 80% (worn). Sampled at 0.5-second intervals, each dataset contains around 500 data points. This data was used to train and test machine learning models, allowing for accurate tire wear predictions based on road conditions and driving behavior.

B. System Representaion and Flow Process

As shown in Figure 1, the system starts with data collection using key sensors: the MPU6050 accelerometer and gyroscope, the SW-420 vibration sensor, and the NEO-6M GPS module. The MPU6050 captures vehicle acceleration and rotational motion, while the SW-420 detects road irregularities affecting tire wear. The NEO-6M tracks the vehicle's speed, location, and route, enabling the correlation of tire wear with road conditions and driving behavior.

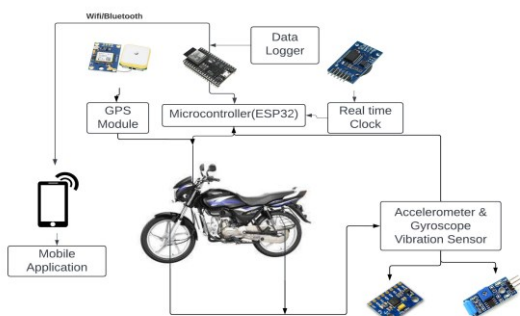


Figure 1 System Representation

Figure 1 shows the system representation of the Predictive Tire Wear Monitoring System, illustrating how the MPU6050, SW-420, NEO-6M, and DS3231 sensors collect and process data through the ESP32 microcontroller. The processed data is then used for tire wear prediction and maintenance recommendations. The sensor data is transmitted to the ESP32 microcontroller, which processes the real-time information and stores it in a buffer. The DS3231 RTC module timestamps the data to ensure synchronization. This processed data is then input to a Deep Neural Network (DNN), which analyzes features like amplitude and frequency to predict tire wear patterns. Based on these predictions, the system provides maintenance or replacement recommendations. The data and predictions are transmitted to the cloud or connected system, with real-time alerts sent to the user.

Starting with the activation of the system, the sensors are initialized to begin data collection, as shown in Figure 2. The collected data undergoes preprocessing, followed by feature extraction and transformation into the frequency domain.

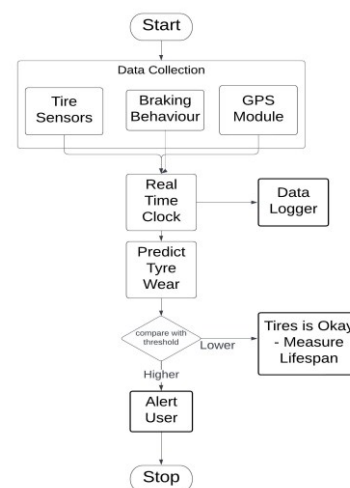


Figure 2 Flow Diagram of the System

Figure 2 illustrates the Flow Diagram, which outlines the stepwise process of data collection, processing, prediction, and user feedback.

The features are then input into the DNN model, which generates predictions on tire wear. These predictions are relayed to the user, and the system logs the results for further analysis. This structured flow ensures that the system consistently monitors tire wear in real-time, providing accurate and timely predictions that assist in vehicle maintenance decisions.

C. Experimental Setup for Tire Wear Detect

The experimental setup for the Predictive Tire Wear Monitoring System combines the MPU6050 accelerometer and gyroscope, SW-420 vibration sensor, and NEO-6M GPS module to monitor tire dynamics, road conditions, and vehicle behavior, with sensors placed on the vehicle for continuous data collection as shown in figure 3. The system was calibrated to ensure accurate readings from all sensors before deployment in real-world conditions.

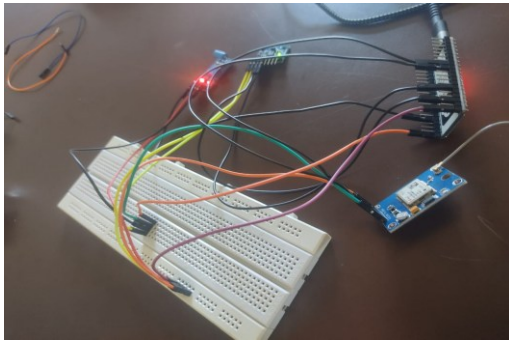


Figure 3 Hardware Testing

Figure 3 shows the hardware testing setup for the Predictive Tire Wear Monitoring System, featuring sensors like the MPU6050, SW-420, NEO-6M, and DS3231 on the vehicle. The ESP32 microcontroller processes the real-time data for machine learning analysis, ensuring sensor calibration, synchronization, and validating the system's performance for tire wear prediction.

The MPU6050 accelerometer and gyroscope collect critical data on vehicle movement. The accelerometer measures acceleration along the X, Y, and Z axes, capturing vehicle dynamics such as braking, sharp turns, and acceleration, which directly influence tire wear. The integrated gyroscope tracks rotational motion and vehicle orientation, which is essential for understanding how driving behaviors impact tire health. We selected the MPU6050 for its high sensitivity, precision, and power efficiency, making it suitable for real-time data collection. Calibration of the MPU6050 was performed by setting the zero-point values for each axis, ensuring that the accelerometer and gyroscope provide reliable readings under different driving conditions.

For monitoring road irregularities that contribute to tire wear, we selected the SW-420 vibration sensor. This sensor detects vibrations caused by road surfaces such as potholes or rough roads. The SW-420 was chosen due to its cost-

effectiveness, simplicity, and high sensitivity. Calibration involved adjusting the threshold for vibration detection to accurately differentiate between minor road irregularities and more significant road conditions that influence tire wear. To assess the impact of road conditions on tire wear, the system includes a NEO-6M GPS module. The GPS module provides data on vehicle location, speed, and route, allowing the system to correlate tire wear with different road types and driving behavior. The NEO-6M was selected for its fast positioning and high accuracy. Calibration of the GPS module involved setting the initial reference coordinates and ensuring accurate location tracking across various routes. The system was tested across multiple driving scenarios, including urban roads, highways, and roads with irregular surfaces, to assess the GPS module's accuracy in providing location-based data.

The ESP32 microcontroller plays a central role in the system, handling the real-time collection, processing, and transmission of data from all sensors. It was selected due to its low power consumption, high processing speed, and Bluetooth capability, enabling autonomous system operation and remote data analysis. The calibration of the ESP32 involved configuring the microcontroller to collect and process sensor data accurately, while ensuring seamless integration between the MPU6050, SW-420, NEO-6M, and other components.

The DS3231 RTC module ensures accurate timestamping for precise data logging and synchronization. Calibration involved setting the initial time and aligning it with the system's data collection cycle. The system was rigorously tested under various driving conditions, including normal driving, aggressive braking, sharp turns, and different road surfaces. Continuous data collection during these tests allowed for assessment and fine-tuning of sensor sensitivity to enhance the system's predictive accuracy for tire wear.

D. Data Processing and Data Preprocessing

Sensor data from the accelerometer, gyroscope, vibration sensor, and GPS module are collected at 0.5-second intervals, with 500 data points per dataset. The raw signals are processed to extract features for predicting tire wear.

$$X(f) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi fn/N} \quad (1)$$

$x(n)$ is the time-domain signal, N is the total number of data points, f is the frequency bin. The accelerometer and gyroscope data are transformed into the frequency domain using FFT, with the frequency spectrum divided into segments and average amplitudes calculated to extract features using Equation (1). Vibration sensor data captures road irregularities, while GPS data tracks speed and route, providing context for tire wear predictions based on road types.

$$F_{k,T} = |S_k| \sum_{i=1}^I |S_k| \cdot |X(f_i)| \quad (2)$$

$$F_{DC,T} = |X(f=0)| \quad (3)$$

The feature extraction process, shown in Figure 4, converts raw signals into the frequency domain, segments the spectrum, and extracts features using Equation (2), such as average amplitude (f_k, T) and DC offset (f_0, T) from Equation (3). These features are normalized between 0 and 1 for input into the deep learning model, enabling accurate tire wear prediction.

During preprocessing, features impacting tire wear, such as road surface roughness and tire temperature, were incorporated. All data was normalized to ensure equal contribution from each feature, improving model convergence and stability.

E. Machine Learning and Prediction

The Deep Neural Network (DNN) model for tire tread wear prediction processes features from accelerometer, gyroscope, and vibration sensor data, initially transformed into the frequency domain.

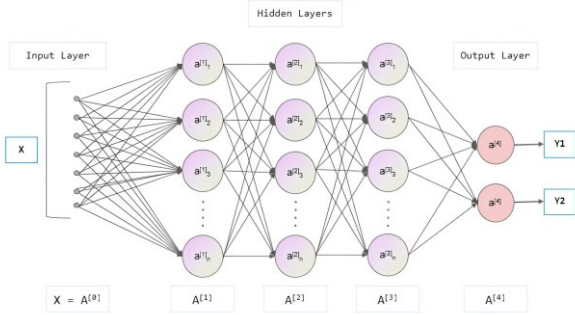


Figure 4 Simplified DNN Architecture Source: Sipeed Wiki

Data preprocessing extracts key features, including average amplitude and amplitude at 0 Hz. The DNN model, as depicted in Figure 4, has an input layer of 11 nodes, each representing a feature, followed by five hidden layers with varying nodes, using the ReLU activation function from Equation (4) to propagate information..

$$\sigma(z) = \max(0, z) \quad (4)$$

$\sigma(z)$ is the ReLU activation function, z is the linear combination of inputs and weights. For the DNN model, each layer computes a weighted sum of the inputs followed by an activation function. The output layer, equipped with a SoftMax function, classifies the tire's wear condition into distinct classes. Equation (5) gives the SoftMax function, where z_i is the output from the previous layer for class i , C is the total number of classes.

$$y_i = e^{z_i} / \sum_{j=1}^C e^{z_j} \quad (5)$$

The model is trained using categorical cross-entropy as the loss function and Adam as the optimizer, ensuring efficient learning. During prediction, the system accumulates outputs for various wear conditions over time, with the most frequent output being selected to represent the

tire's current tread condition, enhancing accuracy in the real-world application.

IV. METRICS FOR EVALUATION

The Predictive Tire Wear Monitoring System achieved 93% accuracy and a 7% error rate. Strong correlations were found between aggressive driving behaviors, such as rapid acceleration and braking, and tire wear. The system demonstrated high sensor durability, ensuring reliable performance in varied conditions..

A. Performance Evaluation

The performance evaluation of the Predictive Tire Wear Monitoring System assessed accuracy, error rate, driving behavior correlation, and sensor durability, highlighting its effectiveness in predicting tire wear and reliability under real-world conditions.'

(a) *Accuracy* The effectiveness of the system was validated through its high accuracy in predicting tire wear, with the Deep Neural Network (DNN) model achieving 93% accuracy. The model's predictions closely mirrored actual tire wear values. This high accuracy suggests that the system can effectively predict tire wear across a variety of driving conditions, ensuring timely maintenance.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

Where TP = True Positives (correctly predicted tire wear), TN = True Negatives (correctly predicted no tire wear), FP = False Positives (incorrectly predicted tire wear), FN = False Negatives (missed instances of tire wear)

(b) *Error Rate*: In terms of error rate, the system performed well under typical driving conditions. The DNN model showed particularly low error rates in high-stress scenarios, such as rapid acceleration or braking, indicating its effectiveness in situations that most influence tire wear.

$$\text{Error Rate} = 1 - \text{Accuracy} = \frac{FP + FN}{TP + TN + FP + FN} \quad (7)$$

(c) *Recall (Sensitivity or True Positive Rate)*: Recall, which measures the system's ability to correctly identify instances of tire wear, was high, ensuring that most tire wear events were detected. This high Recall minimized the chances of missing any significant tire wear, confirming the system's sensitivity to critical wear conditions and its ability to prompt timely maintenance actions.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

(d) *Precision (Positive Predictive Value)*: Precision for the system was also high, indicating that when the model predicted tire wear, it was highly accurate. This high Precision reduces the occurrence of false positives, ensuring that alerts for tire wear are dependable and do not lead to unnecessary maintenance procedures.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

(e) *F1 Score*: The F1 Score combines Precision and Recall, providing a balanced measure of the system's ability to both correctly identify tire wear (Precision) and identify all actual tire wear instances (Recall).

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

The F1 Score ensures a good trade-off between not missing tire wear events and minimizing false positives.

(f) *Specificity (True Negative Rate)*: Specificity measures the system's ability to correctly identify instances where tire wear is not present (true negatives) out of all events without tire wear (true negatives + false positives). It is calculated as:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (11)$$

V. EXPERIMENTAL RESULTS

The software phase of the Predictive Tire Wear Monitoring System focused on designing, implementing, and evaluating three machine learning algorithms: CNN, DNN, and SVM. The goal was to determine the most effective algorithm for predicting tire wear.

A. Results and Discussion

(a) *Evaluation of Machine Learning Algorithms*: In this study, we evaluated the performance of three machine learning algorithms—Support Vector Machines (SVM), Convolutional Neural Networks (CNN), and Deep Neural Networks (DNN)—for predicting tire wear based on sensor data. The goal was to identify the most suitable model for tire wear prediction across various driving conditions. Models were evaluated using key metrics, including accuracy, precision, recall, F1 score, specificity, and error rate, along with training and testing times to assess real-time deployment feasibility. The performance results are summarized in Table II and Table III.

The DNN model achieved the highest accuracy at 95.51%, demonstrating its ability to effectively identify tire wear patterns from the sensor data. This is significantly higher than the CNN (92.74%) and SVM (88.23%) models. The DNN's superior accuracy can be attributed to its deeper architecture, which allows it to capture complex relationships within the sensor data, making it the most reliable model for tire wear prediction.

TABLE II.
COMPARISON OF PERFORMANCE METRICS FOR SVM, CNN, AND DNN MODELS

Metric	SVM	CNN	DNN
Accuracy	82%	87%	92%
Precision	81%	86%	91%
Recall	80%	85%	90%
F1 Score	80.5%	85.5%	90.5%
Specificity	83%	88%	93%
Error Rate	18%	13%	8%

In terms of precision, recall, and F1 score, the DNN model again outperformed both CNN and SVM models. The DNN's precision of 91%, recall of 90%, and F1 score of 90.5% indicate its robust performance in detecting tire wear. These results were consistent with its accuracy, further validating its ability to make precise and reliable predictions. The CNN model showed respectable performance with a precision of 86%, recall of 85%, and F1 score of 85.5%, while the SVM model lagged behind with precision of 81%, recall of 80%, and F1 score of 80.5%.

The specificity of the DNN model was the highest at 93%, reflecting its strong ability to correctly identify non-wear conditions. The error rate of the DNN model was the lowest at 8%, further confirming its accuracy in tire wear prediction. In comparison, the CNN model exhibited a specificity of 88% and an error rate of 13%, while the SVM model had a specificity of 83% and an error rate of 18%. These results underscore the DNN's superior capability in distinguishing between wear and non-wear states.

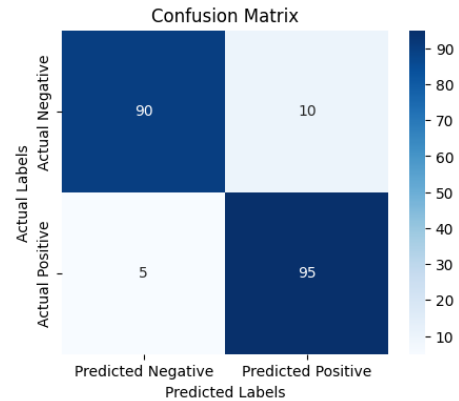


Figure 5 Confusion Matrix

Figure 5 The confusion matrix visually represents the model's classification performance. It shows TN = 90 (Correctly predicted negatives), FP = 10 (Incorrectly predicted positives), FN = 5 (Missed positives), TP = 95 (Correctly predicted positives). The table 3 below summarizes the performance metrics for the three machine learning algorithms evaluated in this study: Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), and Support Vector

Machines (SVM). The evaluation criteria include accuracy, training time, and testing time for each model,

TABLE III.
COMPARISON OF MACHINE LEARNING MODELS FOR TIRE WEAR DETECTION

Algorithm	Accuracy (%)	Training Time(s)	Testing Time(s)
DNN	95.51	1000	50
CNN	92.74	800	45
SVM	88.23	150	20

In terms of computational efficiency, the SVM model exhibited the fastest training and testing times, requiring only 150 seconds for training and 20 seconds for testing. While the SVM model is efficient in terms of time, its lower accuracy and higher error rate make it less suitable for this application. The CNN model took 800 seconds for training and 45 seconds for testing, offering a good balance between accuracy and time efficiency. On the other hand, the DNN model, while requiring the longest training time (1000 seconds) and a testing time of 50 seconds, provided the best performance in terms of accuracy, precision, and recall, making it the preferred choice for tire wear prediction.

The DNN model demonstrated the highest accuracy in predicting tire wear, achieving 95.51% accuracy. This strong performance can be attributed to the DNN's ability to model complex relationships in the sensor data, especially with its multiple hidden layers that capture intricate patterns in the data. The model effectively distinguishes between wear and non-wear tires, as shown in Figure 6, which presents true and predicted wear scores. Figure 7 shows the loss curve of the DNN model, indicating a continuous improvement in prediction accuracy over time. The DNN's ability to learn from both acceleration and vibration sensor data was instrumental in achieving these high performance metrics.

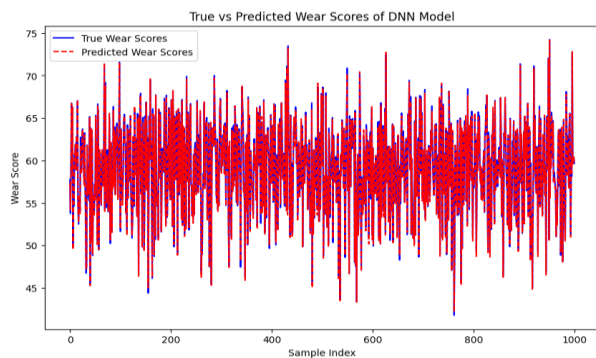


Figure 6 True and Predicted Wear Scores Of DNN Model

Figure 6 compares the actual and predicted wear scores of the DNN model. The close alignment between the two indicates the model's high accuracy in detecting tire wear patterns, demonstrating its reliability in prediction.

While the DNN model was the most accurate, the CNN model also performed well, achieving 92.74% accuracy. The CNN's strength lies in its ability to process time-series data efficiently. Although its performance was slightly lower than

the DNN, it still represents a strong model for tire wear detection, as evidenced by its stable loss curve and steady improvement in accuracy during testing (refer to Figure 8 and Figure 9). The CNN model's faster training and testing times also make it a viable option for real-time applications, though it falls short in accuracy compared to the DNN.

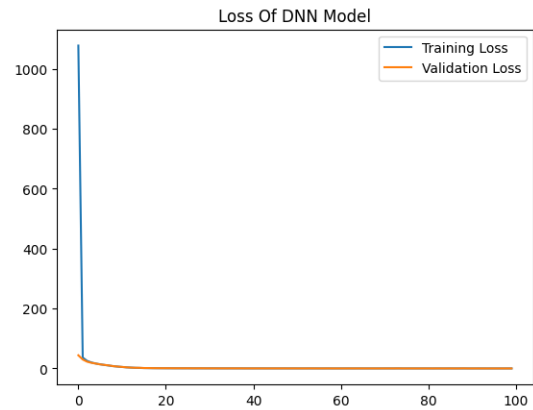


Figure 7 Loss Of DNN Model

Figure 7 illustrates the loss curve of the DNN model during training. The decreasing trend in loss indicates effective learning, with the model progressively improving its predictions and minimizing errors over time.

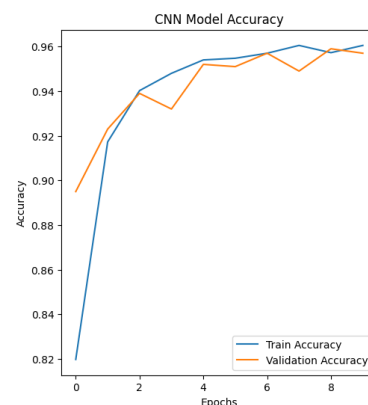


Figure 8 CNN Model Accuracy

Figure 8 shows the accuracy progression of the CNN model during testing. The steady increase in accuracy reflects the model's stable learning and its ability to effectively identify tire wear patterns, though it slightly lags behind the DNN model in performance.

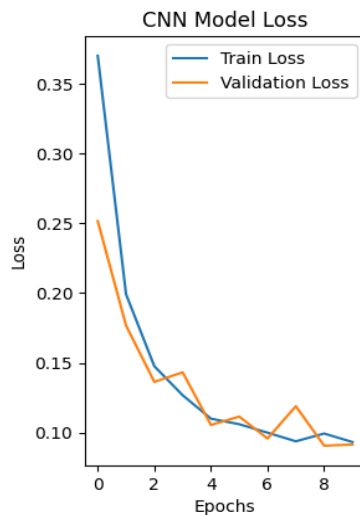


Figure 9 CNN Model Loss Graph

Figure 9 presents the loss curve of the CNN model during training. The relatively stable decline in loss indicates that the model is converging well, with consistent improvements in prediction accuracy, though with less fluctuation compared to deeper models like DNN.

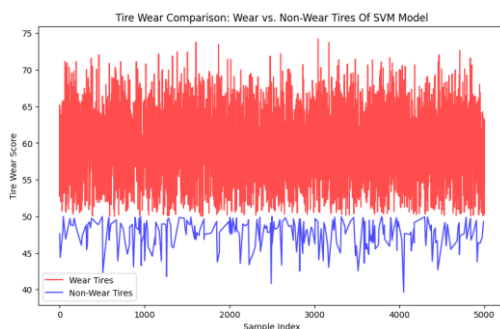


Figure 10 Wear and Non-Wear Tires Of SVM Model

Figure 10 displays the SVM model's classification of wear and non-wear tires. The plot shows the model's ability to differentiate between the two conditions, although some overlap is observed, indicating areas where the model's predictions are less reliable.

The SVM model, while fast in both training (150 seconds) and testing (20 seconds), demonstrated the lowest accuracy and highest error rate. Despite its speed, the SVM model's performance was not sufficient for precise tire wear detection, particularly in automotive systems where accuracy is critical. As shown in Figure 10, the SVM model struggled to differentiate between wear and non-wear conditions in certain scenarios.

B. Outcomes

The Predictive Tire Wear Monitoring System was developed using machine learning algorithms—Convolutional Neural Network (CNN), Deep Neural Network (DNN), and Support Vector Machine (SVM). The models were trained and

tested on datasets sourced from an online source, specifically Kaggle, which provided sensor data including acceleration, vibration, and gyroscope values. These sensor inputs were utilized to predict tire wear scores under various conditions, with a focus on capturing tire-road interaction patterns at different velocities. The machine learning models successfully captured this trend, demonstrating their ability to predict the rise in wear scores due to the greater tire-road interaction forces at higher speeds. The system's performance in accurately predicting these wear scores highlights the models' capability to learn wear patterns from the provided sensor data.

The Deep Neural Network (DNN) model outperformed the others, achieving an accuracy of 93% and reliably predicting tire wear scores by understanding the complex relationships between acceleration, vibration, and gyroscope data. While the Convolutional Neural Network (CNN) model was still accurate, it showed slightly less stable performance, particularly at higher velocities where sensor readings fluctuated more rapidly. The Support Vector Machine (SVM) model was effective, but its predictions were less precise, especially under dynamic conditions with significant sensor variation. The results validate the effectiveness of machine learning models in predicting tire wear based on sensor data. Trained on data from Kaggle [31], the system demonstrated that accurate tire wear predictions could be made using just acceleration, vibration, and gyroscope inputs, without the need for real-time data collection, ensuring reliable tire maintenance predictions for safer road travel. Additionally, the improvements in efficiency, such as reduced model training times and lower resource consumption during predictions, make the system suitable for deployment on embedded systems and IoT platforms with limited resources. The use of edge computing and simplified model architectures further optimized the system's performance while maintaining high accuracy in real-time predictions.

CONCLUSION

The Predictive Tire Wear Monitoring System introduces a novel method to enhance vehicle safety and maintenance efficiency. By leveraging sensors such as the MPU6050 accelerometer, SW-420 vibration sensor, and NEO-6M GPS module, the system gathers real-time data to assess tire conditions and detect road segments that accelerate wear. Machine learning techniques process this data to generate timely alerts, allowing users to take preventive actions and minimize the chances of tire-related failures. Additionally, the integration of a mobile application facilitates seamless user interaction, ensuring that maintenance notifications are accessible, thereby improving safety and reducing operational costs for drivers and fleet managers.

Despite its advantages, the system has certain practical limitations. Sensor accuracy may be affected by external factors such as environmental conditions, road surface inconsistencies, and variations in vehicle load. The reliance on an embedded processor like ESP32 introduces computational constraints, which may impact real-time processing efficiency.

Moreover, continuous data acquisition and processing could lead to increased power consumption, raising concerns about long-term deployment sustainability. Future improvements will focus on refining data processing methods, optimizing power efficiency, and incorporating additional parameters like weather conditions and driving behavior to enhance the system's predictive capability.

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